

三角関数の微分の証明

$$\begin{array}{ll} (\sin x)' = \cos x & (\cos x)' = -\sin x \\ (\tan x)' = \frac{1}{\cos^2 x} & \left(\frac{1}{\tan x}\right)' = -\frac{1}{\sin^2 x} \end{array}$$

(証明)

$$\begin{aligned} (\sin x)' &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos x \sinh - \sin x (1 - \cosh)}{h} \\ &= \lim_{h \rightarrow 0} \left(\cos x \cdot \frac{\sinh}{h} - \sin x \cdot \frac{1 - \cosh}{h^2} \cdot h \right) \\ &= \cos x \cdot 1 - \sin x \cdot \frac{1}{2} \cdot 0 \\ &= \cos x \quad \square \end{aligned}$$

$$\begin{aligned} (\cos x)' &= \left\{ \sin\left(\frac{\pi}{2} - x\right) \right\}' \\ &= \cos\left(\frac{\pi}{2} - x\right) \cdot (-1) \\ &= -\sin x \quad \square \end{aligned}$$

$$\begin{aligned} (\tan x)' &= \left(\frac{\sin x}{\cos x} \right)' \\ &= \frac{\cos x \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} \quad \square \end{aligned}$$

$$\begin{aligned} \left(\frac{1}{\tan x}\right)' &= \left\{ (\tan x)^{-1} \right\}' \\ &= -(\tan x)^{-2} \cdot \frac{1}{\cos^2 x} \\ &= -\frac{1}{\tan^2 x} \cdot \frac{1}{\cos^2 x} \\ &= -\frac{1}{\sin^2 x} \quad \square \end{aligned}$$